PSGR
Krishnammal College for Women

PRODUCTION AND INVENTORY MONITORING DASHBOARD

PROJECT WORK SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF BACHELOR OF COMPUTER SCIENCE WITH COGNITIVE SYSTEMS

Submitted by

POOJAA V. K

22BCC040

Guided by

Dr. (Mrs.) J. Viji Gripsy, M.Sc., M.Phil., Ph.D.

Associate Professor & Head

Department of Computer Science with Cognitive Systems

DEPARTMENT OF COMPUTER SCIENCE WITH COGNITIVE SYSTEMS

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CERTIFICATE

CERTIFICATE

This is to certify that the project work entitled **"Production and Inventory Monitoring Dashboard"** submitted to Bharathiar University in partial fulfilment of the requirement for the award of Degree of the Bachelor of Computer Science with Cognitive Systems is a record of the original work done by **POOJAA V. K (22BCC040),** during her period of study in Department of Computer Science with Cognitive Systems, PSGR Krishnammal College for Women, Coimbatore under my supervision and guidance and her project work has not formed the basis for the award of any Degree/Diploma/Associate/Fellowship or similar title to any candidate of any university.

Forwarded by

Dr. (Mrs.) J. Viji Gripsy **Dr. (Mrs.) J. Viji Gripsy**

Faculty Guide Head of the Department

Submitted for the final examination held on _____

Internal examiner **External examiner**

DECLARATION

DECLARATION

I hereby declare that this project work entitled **"Production and Inventory Monitoring Dashboard"** submitted for award of the Degree of Bachelor of Computer Science with Cognitive Systems is a record of the original work done by **POOJAA V. K (22BCC040),** under the supervision and guidance of **Dr. (Mrs.) J. Viji Gripsy, Associate Professor & Head,** Department of Computer Science with Cognitive Systems, PSGR Krishnammal College for Women, Coimbatore and this project work has not formed the basis for the award of any Degree or similar title to any candidate of any university.

Place: Coimbatore **POOJAA V. K**

Date: **22BCC040**

Endorsed by

Place: Coimbatore **Dr. (Mrs.) J. Viji Gripsy**

Date: (Faculty Guide)

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"Success is to be measured not so much by the position that one has reached in life but as by the obstacle which he had to overcome while trying to succeed."

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SYNOPSIS

SYNOPSIS

The “Production and Inventory Monitoring Dashboard” project is designed to provide a clear and dynamic view of production processes and inventory management using Microsoft Power BI. By integrating data from the “Miniwee 2024” dataset, which contains 5,000 records across 42 attributes, the dashboard offers valuable insights into key areas like product details, customer demographics, sales performance and logistics. Important variables such as product categories, selling prices, profit margins, stock levels and delivery times form the core of this analysis, helping to evaluate operational efficiency and financial health.

The main objective of this project is to simplify and enhance decision-making processes by providing insights into production and inventory management. This dashboard identifies the most profitable products, pinpoints areas where costs can be optimized and provides a deeper understanding of customer behaviour. Through profitability and cost analysis, the dashboard highlights high-margin products and areas where operational costs can be reduced. Furthermore, it provides a comprehensive understanding of customer purchasing patterns and engagement across various marketing channels, helping businesses implement targeted strategies to drive sales and improve customer retention.

The dashboard includes several interactive elements designed to enhance usability and provide a comprehensive understanding of business operations. KPI cards highlight core business metrics such as inventory turnover, stock levels, profit margins and delivery times. Trend visualization charts track production output and inventory movement over time, helping to identify patterns and anomalies. This interactivity ensures that stakeholders can easily interpret complex data and derive actionable insights to drive operational efficiency and business growth. By offering insights, the dashboard improves transparency across production and inventory processes. It enables businesses to identify bottlenecks, reduce inefficiencies and optimize resource allocation. This data-driven approach supports better inventory control, reduces the risk of stockouts or overstocking and enhances overall supply chain efficiency.

INTRODUCTION

1. INTRODUCTION

In today's fast-paced industrial environment, effective production and inventory management is crucial for maintaining operational efficiency and sustaining business growth. Organizations are constantly seeking advanced solutions to monitor and optimize these processes in . Traditional methods of tracking production and inventory often lead to inefficiencies, delays and inaccuracies, which can significantly impact a company's profitability and customer satisfaction. With the increasing availability of big data and advanced analytics, there is a growing demand for intelligent systems that provide insights to drive informed decision-making. This project, titled " Production and Inventory Monitoring Dashboard," addresses these challenges by offering an interactive, data-driven solution using Microsoft Power BI. The dashboard is designed to deliver visibility into key aspects of production and inventory processes, including product performance, customer demographics, sales analysis and supply chain logistics.

The primary goal of this project is to create a user-friendly dashboard that assists businesses in monitoring production output, managing inventory levels and identifying areas for improvement. By analysing crucial variables such as product categories, selling prices, profit margins, stock levels and delivery times, the dashboard facilitates a deeper understanding of operational efficiency. Through interactive visualizations, users can track performance metrics, identify trends and gain insights into customer purchasing behaviours. This holistic approach enables businesses to reduce operational costs, improve inventory accuracy and respond quickly to market demands.

By leveraging the advanced analytics capabilities of Power BI, the dashboard not only provides monitoring but also allows for the analysis of historical data to uncover patterns and anomalies. This comprehensive view supports better forecasting, resource allocation and inventory control. Furthermore, the dashboard promotes data transparency and ensures that stakeholders at all levels have access to the information they need to make timely and strategic decisions.

1.1 Organizational Profile

Miniwee is an online clothing store specializing in premium baby and maternity products. Established by Ashwin Kumar T R and Muthu Hari Prasath Anguraj, the company was founded with the vision of providing high-quality, safe and stylish baby essentials to modern parents. Miniwee's mission is to empower parents by offering thoughtfully curated baby products that prioritize safety, comfort and sustainability.

The company takes pride in producing high-quality baby products using 100% organic materials, including muslin fabric known for its breathability, softness and durability. By prioritizing organic and eco-friendly manufacturing processes, Miniwee emphasizes sustainability while delivering products that meet the highest safety and quality standards. Each product is crafted with meticulous attention to detail, reflecting a perfect combination of modern design and traditional craftsmanship.

Miniwee’s diverse product range includes baby apparel, accessories and maternity essentials that cater to the evolving needs of modern parents. The company supports local artisans and promotes sustainable production practices, contributing to environmental responsibility and community welfare. This dedication to ethical business practices ensures that Miniwee’s products are not only safe and stylish but also environmentally conscious.

Through innovation and customer-centric approaches, Miniwee has established itself as a trusted brand for parents seeking premium, reliable and sustainable baby products. The company's commitment to quality, transparency and customer satisfaction has helped it build a strong presence in the market. By combining advanced manufacturing techniques with a deep understanding of parenting needs, Miniwee continues to deliver products that make parenting easier and more enjoyable.

1.2 Project Overview

The “Production and Inventory Monitoring Dashboard” project aims to enhance manufacturing operations by providing insights into production and inventory processes. This dashboard, developed using Microsoft Power BI, is designed to track and analyse key performance indicators (KPIs) such as production efficiency, inventory levels, sales performance and supply chain dynamics.

By leveraging data from the “Miniwee 2024” dataset, the dashboard offers monitoring capabilities that help identify inefficiencies, predict inventory needs and support data-driven decision-making. It provides stakeholders with a centralized view of critical operations, allowing for better resource allocation, improved forecasting and enhanced operational efficiency. The dashboard is interactive and user-friendly, offering dynamic filters and visualization tools to explore data from multiple angles. This enables managers to quickly identify production bottlenecks, track product movement and optimize supply chain workflows. The system is capable of handling large datasets and updating information in , ensuring accuracy and responsiveness.

The project’s primary objective is to provide a comprehensive and scalable solution for monitoring and improving manufacturing performance. By integrating advanced data analytics with visualization, the dashboard empowers decision-makers to streamline operations, reduce waste and improve overall productivity. This project represents a significant step toward digital transformation, enabling Miniwee to maintain its commitment to quality and operational excellence in a competitive marketplace.

1.3 System Environment

The system environment for the Production and Inventory Monitoring Dashboard includes both hardware and software components that facilitate seamless data collection, processing and visualization.

1.3.1 Hardware Requirements

- Processor - Intel Core i7 or equivalent (for optimal data processing speed)
- Memory (RAM) - Minimum 16 GB (to handle large datasets efficiently)
- Storage - At least 1 TB SSD (for faster data retrieval and storage)
- Graphics Card - Dedicated GPU (for rendering complex visualizations)
- Network Infrastructure - High-speed LAN/WAN connectivity
- Servers – Cloud based or on premise servers to manage data flow and storage

1.3.2 Software Requirements

- Operating System - Windows 10/11 (64-bit) or Windows Server (for hosting the dashboard)
- Microsoft Power BI - For building and managing interactive dashboards
- ETL Tools - Power Query (for data extraction, transformation and loading)
- Security Software - Data encryption and user authentication mechanisms

These hardware and software components work together to ensure accurate data flow, high-speed processing and visualization. The system environment is designed to be scalable and adaptable, allowing Miniwee to monitor its production and inventory processes effectively and make data-driven decisions.

1.4 Software Tools & Features

Microsoft Power BI is a powerful business intelligence tool that allows users to transform raw data into interactive and insightful visualizations. It is a comprehensive analytics platform that facilitates data modelling, visualization and reporting, making it an ideal choice for production and inventory monitoring. Power BI offers seamless integration with various data sources such as SQL databases, Excel files and cloud-based storage. This flexibility allows the dashboard to collect and process data from different operational units. Its intuitive drag-and-drop interface makes it easy to create and customize visual reports without requiring extensive coding knowledge.

A key feature of Power BI is its ability to deliver updates, enabling stakeholders to monitor production metrics and inventory levels instantly. This feature enhances decision-making by providing accurate and up-to-date information. Additionally, Power BI supports advanced data modelling capabilities, allowing for the creation of complex relationships between datasets, which is essential for analysing production workflows and inventory tracking. Another significant aspect of Power BI is its robust security framework.

It offers role-based access control, data encryption and compliance with industry standards, ensuring that sensitive business data remains protected. Users can also share reports across the organization using Power BI Service, promoting collaborative decision-making.

The tool's scalability ensures that it can handle large datasets while maintaining high performance. With the Power BI Mobile App, decision-makers can access reports and dashboards on the go, facilitating oversight and swift responses to production challenges. Microsoft Power BI is a comprehensive and user-friendly tool that plays a central role in the Production and Inventory Monitoring Dashboard. Its advanced analytics, capabilities and strong security features make it indispensable for tracking, analysing and optimizing manufacturing processes.

SYSTEM STUDY AND ANALYSIS

2. SYSTEM STUDY AND ANALYSIS

The System Study and Analysis phase plays a vital role in the successful development of the Production and Inventory Monitoring Dashboard. It involves a detailed examination of Miniwee's existing production and inventory management processes to identify inefficiencies and establish system requirements. The primary goal of this phase is to understand the limitations of the current manual system, recognize technological gaps and design an automated solution for monitoring. The study reveals the need for improved data accuracy, faster reporting and seamless integration across different operational units. By implementing a centralized dashboard, Miniwee can transition from manual data collection to an automated process that ensures visibility, enhances decision-making and improves operational transparency. This system is designed to provide live tracking of production and inventory levels, enabling management to respond swiftly to fluctuations and operational challenges. The insights gained from this analysis form the foundation for a robust, scalable and secure monitoring solution that not only streamlines workflows but also supports Miniwee's commitment to innovation and operational excellence.

2.1 Existing System

Miniwee's current production and inventory management system primarily relies on manual data entry and periodic reporting. Data is collected through paper records, spreadsheets and manual logs maintained by staff across production lines and inventory checkpoints. This manual process involves recording production quantities, tracking raw material consumption and updating stock levels by hand. Periodic reports are generated weekly or monthly, which delays the availability of critical insights needed for quick decision-making.

This approach has several drawbacks, including data inaccuracies due to human error, time-consuming processes and difficulty in tracking production and inventory movements. Manual data entry also limits the ability to analyse historical trends effectively, as data must be manually compiled and validated before being used for forecasting. Additionally, the absence of automated alerts means that stock shortages, production delays, or equipment inefficiencies often go unnoticed until the next reporting cycle, which can disrupt operations.

The reliance on spreadsheets further complicates collaboration across departments, as version control issues and fragmented data sharing create inconsistencies. The manual system is labour-intensive, error-prone and lacks the agility required to keep up with dynamic production environments. The analysis highlighted the urgent need for a more efficient, automated solution to enhance data accuracy, streamline processes and provide operational insights.

2.2 Proposed System

The proposed Production and Inventory Monitoring Dashboard leverages Microsoft Power BI to automate and enhance Miniwee's production and inventory management. Power BI's advanced analytics and visualization capabilities transform the manual, error-prone processes into a dynamic, monitoring system. This system provides accurate, up-to-the-minute insights that enable faster decision-making and operational efficiency.

By integrating Power BI with Miniwee's production data sources, tracking of production outputs, inventory levels and operational performance becomes possible. The dashboard consolidates data from multiple systems, offering a centralized platform for monitoring and analysing key performance indicators (KPIs). This reduces the need for manual data entry and minimizes human error. Automated data processing improves accuracy while reducing the time required to generate critical business insights.

One of the most significant advantages of incorporating Power BI is its ability to provide interactive, user-friendly dashboards. Management can easily access and interpret complex data through visual charts, graphs and reports. This ensures that decision-makers can quickly identify trends, detect bottlenecks and take proactive measures to optimize operations. Moreover, automated alerts within the dashboard notify stakeholders of critical deviations, such as low inventory levels or production delays, enabling timely interventions.

The Power BI solution is also scalable, allowing Miniwee to expand its monitoring capabilities as the business grows. It supports seamless data integration across multiple departments, fostering better collaboration and ensuring consistency. Additionally, the system ensures data security through controlled access, allowing only authorized personnel to view or modify sensitive information.

The proposed system using Power BI transforms Miniwee's production and inventory monitoring into a faster, more accurate and efficient process. It addresses the shortcomings of the manual system by providing insights, improving data reliability and enhancing decision-making capabilities. This technological advancement positions Miniwee to respond swiftly to operational challenges and maintain its commitment to innovation and excellence.

2.3 Dataflow Diagram

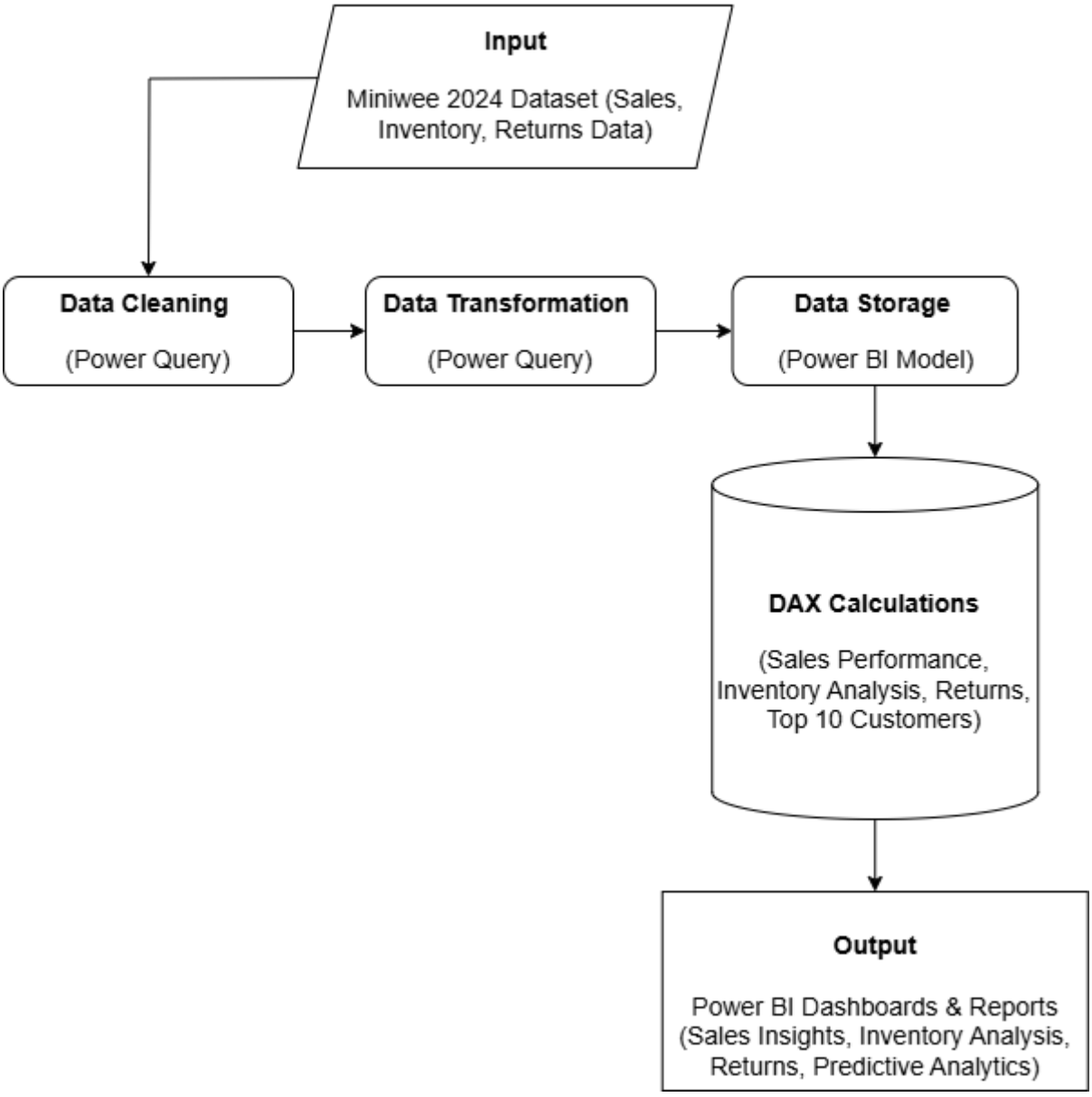

Figure 1 Power BI Process Overview

SYSTEM DESIGN AND METHODOLOGY

3. SYSTEM DESIGN AND METHODOLOGY

The system design and methodology for the Production and Inventory Monitoring Dashboard at Miniwee focus on automating data collection, improving data accuracy and enabling decision-making. The system is built using Microsoft Power BI, integrating data from various operational sources to provide a centralized platform for monitoring and analysis.

The system architecture follows a structured workflow, beginning with data acquisition. Production and inventory data are provided in Excel format, capturing essential information such as raw material consumption, production outputs and stock levels. This data is uploaded into Power BI, where it undergoes cleaning and transformation to ensure consistency and accuracy. The data processing stage involves analysing uploaded data to derive meaningful insights. Key metrics such as production efficiency, inventory turnover and deviation alerts are calculated. This stage ensures the identification of trends and operational bottlenecks through advanced data modelling and analytics capabilities in Power BI.

In the dashboard creation phase, interactive dashboards are designed to present visualizations. These dashboards are tailored to Miniwee's requirements, displaying key performance indicators (KPIs) and providing automated alerts for critical deviations. Users can monitor live production status, track inventory levels and generate on-demand reports for in-depth analysis.

The methodology employs an iterative, user-centered design approach to ensure the system meets operational needs. Regular feedback loops with stakeholders guide the development process, allowing continuous refinement of dashboards and analytics. Power BI's data refresh capabilities ensure that the system updates in , providing accurate and timely insights. The system facilitates data-driven decision-making by providing actionable intelligence. Management can quickly identify inefficiencies, optimize inventory and respond to operational challenges. This automated, monitoring system transforms Miniwee's manual processes into a dynamic solution that enhances operational efficiency and supports strategic decision-making.

3.1 Data Sources and Collection

The data used for the Production and Inventory Monitoring Dashboard is sourced from Miniwee's 2024 operational data, which provides comprehensive information on the company's production and inventory processes. This dataset is crucial for creating a system that enables accurate, monitoring and analysis. The data is collected from multiple touchpoints across Miniwee's production facilities and inventory checkpoints, ensuring a detailed and holistic view of the entire operational workflow.

The primary data source is provided in Excel format, which serves as a central repository for data collection. This dataset includes information on production outputs, raw material usage, inventory levels and operational metrics. Data is collected systematically from various departments, including production lines, inventory management and supply chain operations. The key categories of collected data are:

- Production Data - This includes daily production volumes, product categories, shift-wise outputs and operational timelines. It provides valuable insights into production efficiency and helps track performance against planned targets
- Inventory Data - This data encompasses raw material consumption, current stock levels and finished goods. It helps in monitoring inventory turnover, identifying stock shortages and optimizing supply chain processes
- Operational Metrics - Information on production delays, downtime and quality control reports is collected to identify bottlenecks and areas for improvement
- Historical Data - Past production records, inventory trends and historical performance metrics are gathered to enable forecasting and trend analysis

The data collection process involves manual recording by staff at production sites and inventory checkpoints. This information is regularly compiled into Excel sheets, which are subsequently imported into Microsoft Power BI. The system is designed to support automated data extraction from these sheets, ensuring continuous updates and reducing manual intervention.

Within Power BI, the data undergoes a cleaning and transformation process to ensure accuracy and consistency. This involves removing duplicates, handling missing values and standardizing data formats. Power BI's advanced ETL (Extract, Transform, Load) capabilities facilitate efficient data integration, ensuring that the dashboard reflects operational conditions.

The system is capable of synchronization, allowing automated updates as new data becomes available. This feature ensures that the dashboard always reflects the most current information, enabling management to make informed decisions promptly. The comprehensive data collection process ensures that every aspect of Miniwee's production and inventory operations is captured, analysed and visualized for enhanced operational efficiency and decision-making.

3.2 Data Preparation and Cleaning

The Data Preparation and Cleaning process is a critical phase in ensuring the accuracy, consistency and usability of the Miniwee 2024 production and inventory dataset. This phase involves transforming raw data into a structured format suitable for analysis within Microsoft Power BI, enabling accurate monitoring and data-driven decision-making. Given that Miniwee's data is manually collected in Excel format from multiple operational touchpoints, systematic cleaning and preprocessing are essential to mitigate errors and inconsistencies.

3.2.1 Data Import and Integration

The process begins with importing the Miniwee 2024 dataset into Power BI using Power Query Editor. The dataset comprises production quantities, raw material consumption, inventory levels and other operational metrics from various Excel sheets. To ensure a unified view, data from different sources is merged and integrated. Power BI's advanced data connectors allow seamless extraction from multiple Excel files, ensuring a comprehensive dataset. This integration facilitates synchronization and updates, crucial for maintaining accurate insights across the production and inventory processes.

3.2.2 Handling Missing and Incomplete Data

Missing data is a common issue in manually recorded datasets and can significantly impact analytical accuracy. To address this, missing values are systematically identified and handled using suitable techniques. Numerical fields such as production outputs and inventory levels are filled using statistical methods like mean imputation or linear interpolation when patterns allow. In cases where values cannot be estimated reliably, missing records are flagged for manual review. This ensures that no critical information is overlooked, maintaining the integrity of the analysis.

3.2.3 Standardization and Data Formatting

Inconsistent data formats create barriers to accurate reporting. To resolve this, all data points are standardized. This involves aligning date formats, ensuring uniform units of measurement and applying consistent product and category labels. Text fields are cleaned by removing extra spaces, correcting spelling errors and standardizing terminology across the dataset. Additionally, duplicate records are detected and removed to prevent data redundancy and erroneous calculations.

3.2.4 Error Detection and Anomaly Handling

Inaccuracies such as outliers and typographical errors are identified through data validation rules within Power BI. Statistical techniques like Z-score analysis are applied to detect extreme values, which are either corrected or excluded based on contextual understanding. Automated checks flag unusual patterns (e.g., sudden inventory spikes), ensuring that only verified and meaningful data is used.

3.2.5 Data Transformation and Enrichment

The cleaned dataset undergoes transformation to enhance analytical depth. This includes splitting complex fields, merging datasets and creating calculated columns for key performance metrics such as inventory turnover rates, production efficiency and stock deviations. These transformations support deeper analysis and provide dynamic, interactive insights.

3.3 Data Modelling in Power BI

Data modelling is the foundation of creating an efficient and accurate Power BI dashboard. For the " Production and Inventory Monitoring Dashboard" at Miniwee, the data modelling process involves organizing and structuring the Miniwee 2024 dataset to ensure seamless integration, accurate calculations and optimized performance. This structured model enables the dashboard to present insights into production quantities, inventory levels and operational efficiency.

In this project, data is collected from Miniwee 2024, which includes key operational metrics such as production output, raw material consumption, inventory balances and time-based performance records. The first step in data modelling involves importing this data into Power BI. Using the Power Query Editor, the raw data is transformed and cleaned to remove inconsistencies, missing values and formatting issues. This ensures that the data is accurate, well-structured and ready for analysis.

The dashboard uses a star schema model to optimize performance and streamline queries. In this schema, fact tables contain quantitative data (e.g., production output, inventory movement), while dimension tables hold descriptive information (e.g., product categories, time frames and production lines). This design simplifies complex queries by reducing data duplication and ensuring faster calculations.

Relationships are created between tables using unique identifiers like Product ID, Batch Number and Date. These relationships are predominantly one-to-many, enabling the dashboard to filter and aggregate data dynamically. For example, the Production Fact Table is linked to a Date Dimension Table to track production trends over time. Power BI's data modelling interface allows defining active and inactive relationships, which can be utilized through DAX (Data Analysis Expressions) to perform advanced calculations.

The Miniwee dashboard also leverages calculated columns and calculated tables where necessary. For instance, custom columns compute inventory balances by subtracting stock out from stock in, providing immediate visibility into available inventory. This calculation enables tracking and minimizes manual intervention. Effective data modelling not only improves performance but also ensures that insights are accurate and actionable.

By adopting best practices such as using a star schema, optimizing relationships and minimizing the number of calculated columns, the Miniwee dashboard provides a

comprehensive, view of production and inventory. This approach enhances decision-making, reduces errors and ensures that the system remains scalable as Miniwee's operational data grows.

3.4 Dashboard Designing Principles

Designing an effective Power BI dashboard for the " Production and Inventory Monitoring" system involves creating a user-friendly interface that provides clear, actionable insights. The dashboard is designed to meet the specific needs of Miniwee's decision-makers by offering an intuitive layout, data updates and interactive exploration capabilities. The primary goal is to ensure that production and inventory metrics are presented clearly, enabling swift decision-making.

One of the core principles guiding the design is a user-centric approach. The dashboard is tailored to provide Miniwee's managers and operators with immediate access to critical metrics such as total production output, current inventory levels and efficiency rates. Each visualization is chosen to convey complex data in a simple and understandable format, ensuring stakeholders can monitor and interpret the information effectively.

A clear information hierarchy is maintained throughout the dashboard. At the top level, KPI cards display key metrics, allowing users to view essential data at a glance. Below this, supporting charts and tables provide deeper insights into production trends and inventory movements. Interactive filters and slicers allow users to drill down into specific product categories, time periods, or production lines, enhancing the dashboard's flexibility.

Consistency in design is achieved by adhering to standardized formatting across all pages. This includes using a uniform colour scheme aligned with Miniwee's branding and ensuring consistent font styles and chart types. Labels and legends are clear and descriptive, reducing ambiguity and improving the overall user experience. To facilitate interactive exploration, the dashboard includes slicers, cross-filtering and drill-through options. For example, users can select a specific month to view corresponding production data or focus on a particular product category to analyse inventory levels.

Performance optimization is also a key consideration. The dashboard is designed to handle large datasets without compromising speed. Techniques such as data aggregation, optimized relationships and the use of DAX measures are employed to enhance performance. By following these dashboard designing principles, Miniwee's team gains a , accurate and visually cohesive system. This ensures that critical decisions regarding production planning and inventory management can be made confidently and efficiently.

3.5 Measures, DAX Functions and Calculation

In the " Production and Inventory Monitoring Dashboard" for Miniwee, DAX (Data Analysis Expressions) is a powerful formula language used in Power BI to perform advanced calculations and create dynamic measures. These measures help in transforming raw data into actionable insights, enabling better decision-making and operational efficiency. Below are the key DAX measures used in the Miniwee dashboard, along with their significance and how they contribute to monitoring production and inventory.

- **Actual Earning**

This measure calculates the actual revenue generated after deducting total expenses. It provides a clear picture of the business's profitability. This calculation is essential for understanding net profit and analysing financial health.

Actual Earning = [Total Sales] - [Total Expense]

- **Average Orders per Customer**

This measure calculates the average number of orders placed by each unique customer. It helps in analysing customer engagement and purchase frequency. By dividing the total number of orders by the distinct customer count, this measure provides valuable insights into customer retention and purchasing behaviour.

Average Orders per Customer = DIVIDE ([Total Orders], [Unique Customers], 0)

- **Average Product Rating**

This measure calculates the average rating of all products sold, which reflects customer satisfaction and product quality. Tracking product ratings helps identify top-performing products and areas for improvement.

Average Product Rating = AVERAGE ('Insights'[Product Rating])

- **Late Delivery Count**

This measure counts how many orders were delivered after a specific time frame, indicating delivery performance.

Late Delivery Count = CALCULATE (COUNT ('Insights'[Order ID]), 'Insights'[Delivery Time (Days)] > 3)

- **Low Stock Products**

This measure identifies how many products have low stock, ensuring proactive inventory management. It helps avoid stockouts and ensures products are available to meet customer demand.

Low Stock Products = CALCULATE (COUNT ('Insights'[Product ID]), 'Insights'[Stock Availability] = "Low Stock")

- **On-Time Delivery Count**

This measure counts orders delivered within the expected time frame. Tracking on-time deliveries helps evaluate supplier efficiency and delivery reliability.

On Time Delivery Count = CALCULATE (COUNT ('Insights'[Order ID]), 'Insights'[Delivery Time (Days)] <= 3)

- **Out of Stock Products**

This measure counts how many products are currently unavailable. Identifying out-of-stock items supports better inventory forecasting and order fulfilment.

Out of Stock Products = CALCULATE (COUNT ('Insights'[Product ID]), 'Insights'[Stock Availability] = "Out of Stock")

- **Repeat Customers**

This measure identifies customers who have made more than one purchase. Analysing repeat customers helps assess customer loyalty and the effectiveness of marketing efforts.

Repeat Customers = CALCULATE (DISTINCTCOUNT ('Insights'[Customer ID]), 'Insights'[Order Frequency] > 1)

- **Return Cost Impact**

This measure calculates the financial impact of returned products on overall profitability. Understanding return costs helps optimize return policies and product quality.

Return Cost Impact = SUM ('Insights'[Return Quantity]) * AVERAGE ('Insights'[Cost Price])

- **Return Rate**

This measure calculates the percentage of returned products relative to total orders. Monitoring return rates helps evaluate product quality and customer satisfaction.

Return Rate = DIVIDE (SUM ('Insights'[Return Quantity]), SUM ('Insights'[Total Orders]), 0)

- **Stock Turnover Ratio**

This measure indicates how quickly inventory is sold and replenished. A high turnover ratio indicates strong sales and efficient inventory management.

Stock Turnover Ratio = DIVIDE ([Cost of Goods Sold], [Average Inventory], 0)

- **Total Expense**

This measure calculates the total expenses incurred during the reporting period. Tracking expenses is crucial for maintaining financial control and optimizing costs.

Total Expense = SUM ('Insights'[Total Expense])

- **Total Profit**

This measure calculates the total profit by subtracting expenses from total revenue. Monitoring profit helps in assessing business performance and growth.

Total Profit = [Total Sales] - [Total Expense]

- **Total Sales**

This measure sums the total revenue from all product sales. It is a primary KPI for tracking business growth and sales performance.

Total Sales = SUM ('Insights'[Selling Price])

- **Trendline (Linear Regression)**

This measure calculates the slope to analyse sales trends over time. Using a trendline helps in forecasting future sales and identifying growth patterns.

Slope = SLOPE ('Insights'[Selling Price], 'Insights'[Order Date])

- **Unique Customers**

This measure counts the distinct number of customers in the dataset. Tracking unique customers provides valuable insights into customer acquisition and retention.

Unique Customers = DISTINCTCOUNT ('Insights'[Customer ID])

3.6 Data Visualization Techniques

Data visualization plays a crucial role in enhancing decision-making through the Production and Inventory Monitoring Dashboard. By transforming raw data into visually compelling insights, stakeholders can quickly interpret trends, identify inefficiencies and monitor key performance indicators (KPIs). In this project, Power BI was used to develop dynamic, interactive visualizations that track sales, inventory, supplier performance and customer behaviour.

3.6.1 Visualization Techniques Used

- **Bar and Column Charts**

Bar and column charts are used to compare categorical data. In the dashboard, bar charts display monthly sales and profit trends, allowing users to identify seasonal fluctuations and high-performing months. Additionally, column charts present supplier delivery performance and order fulfilment rates, facilitating the analysis of supplier efficiency. These visuals help compare metrics like return costs, shipping costs and profit margins across product categories.

- **Line Charts**

Line charts are effective for showing continuous data over time. This dashboard leverages line charts to track revenue vs. expenses and monthly customer acquisition trends. By visualizing patterns like the impact of promotions on sales, stakeholders can observe performance changes and adjust strategies to maintain profitability.

- **Pie and Donut Charts**

Pie and donut charts summarize proportional data effectively. This visualization technique was applied to illustrate procurement spend distribution by supplier and customer subscription status. These charts provide a quick overview of which suppliers contribute the most to procurement expenses and the percentage of customers who have subscribed to specific services.

- **Stacked and Clustered Charts**

The dashboard incorporates stacked bar charts to analyse the impact of returns on costs across product categories. Clustered bar charts are used to compare stock availability, showing in-stock, low-stock and out-of-stock products across different categories. This allows quick detection of supply chain vulnerabilities and helps forecast future procurement needs.

- **Card Visuals**

Card visuals provide a clear, at-a-glance summary of key metrics. This dashboard uses card visuals to highlight critical KPIs such as total profit, unique customers, average product rating and stock turnover ratio. These cards deliver quick insights without requiring detailed analysis.

- **Maps and Geographic Visuals**

Geographic data is visualized using map charts to track shipping costs across regions. This provides spatial insights into how delivery costs vary by location, assisting in optimizing logistics and reducing expenses.

- **Matrix and Tables**

Matrix visuals are used to present supplier cost vs. selling price trends, offering a detailed breakdown of financials. Tables are implemented for detailed lists of best-selling and least-profitable products, providing granular visibility for stakeholders interested in specific inventory items.

3.6.2 Interactivity and User Experience

The dashboard is designed to be interactive, allowing users to filter data by date range, product category, warehouse and supplier. Power BI slicers are used to provide these filter options, enabling tailored views that help stakeholders focus on specific areas. Drill-through and tooltips offer in-depth exploration of underlying data, providing a multi-layered analysis approach. The Production and Inventory Monitoring Dashboard leverages diverse visualization techniques to present comprehensive, actionable insights.

3.7 Input & Output Screens

3.7.1 Input Screens

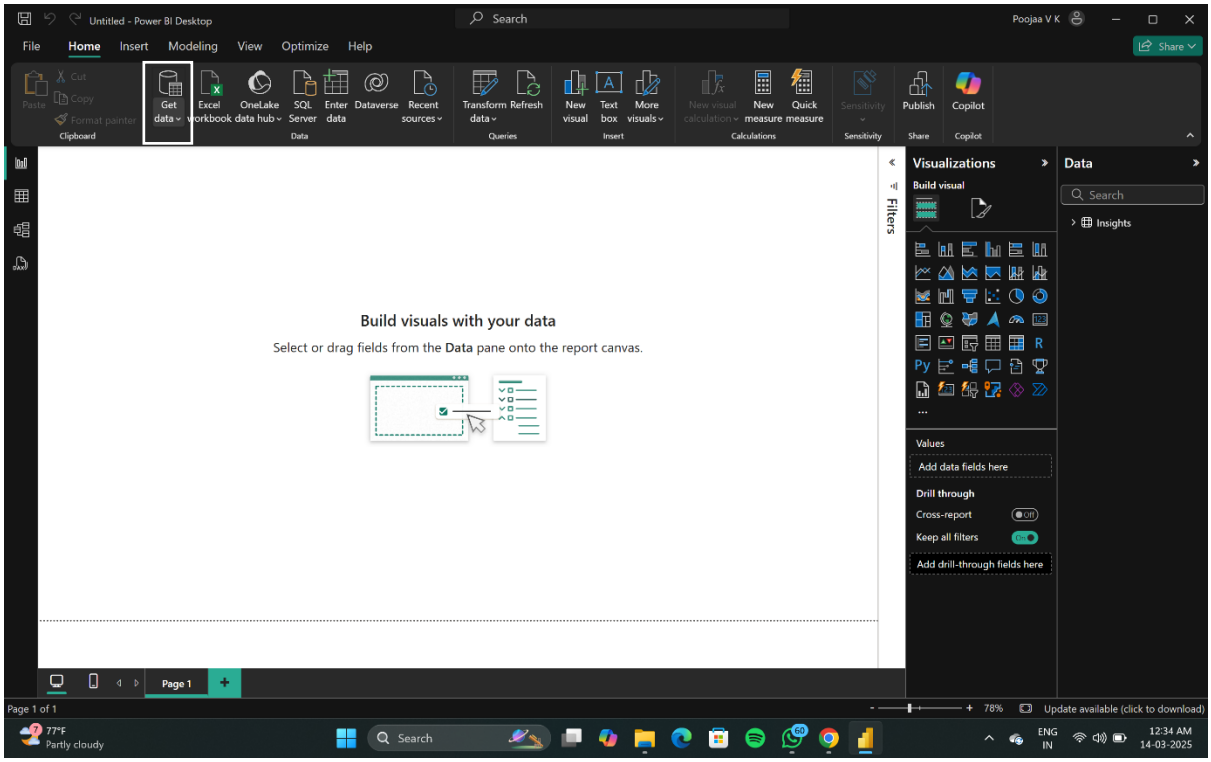

Figure 2 Data Import Interface

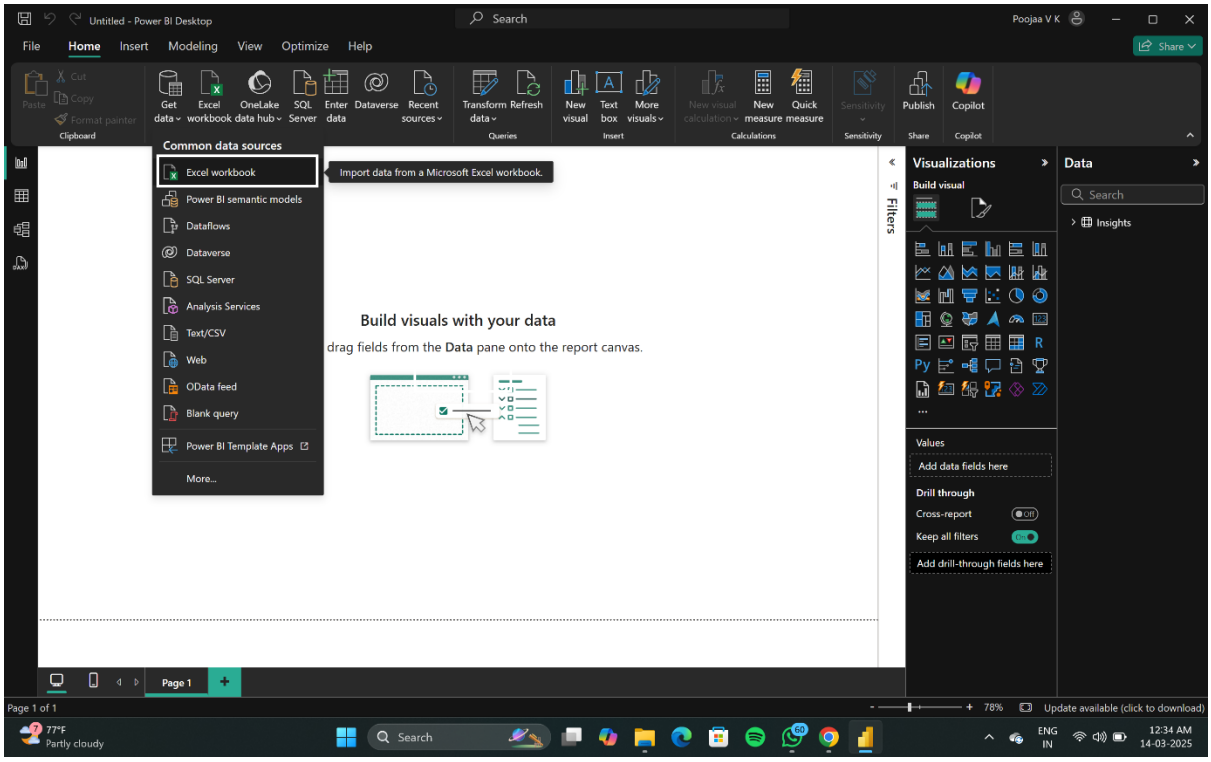

Figure 3 Selecting Data Source (Excel)

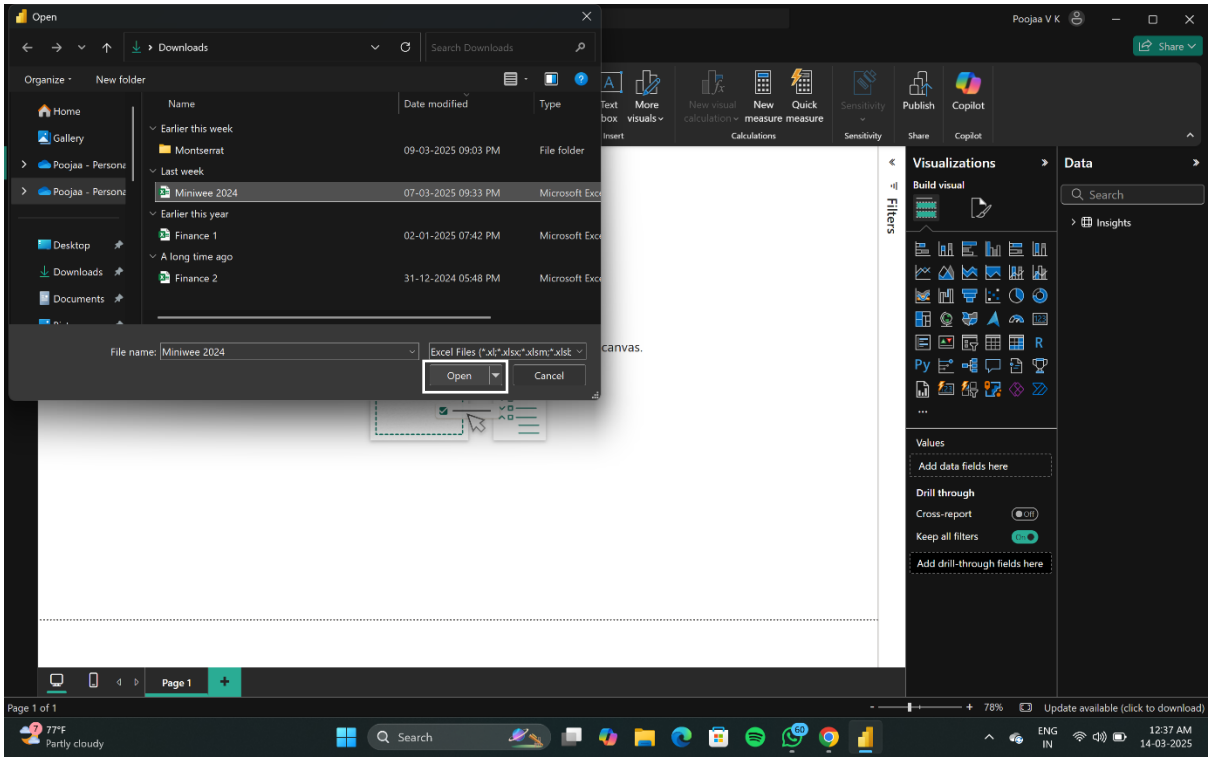

Figure 4 Opening Miniwee 2024 Dataset

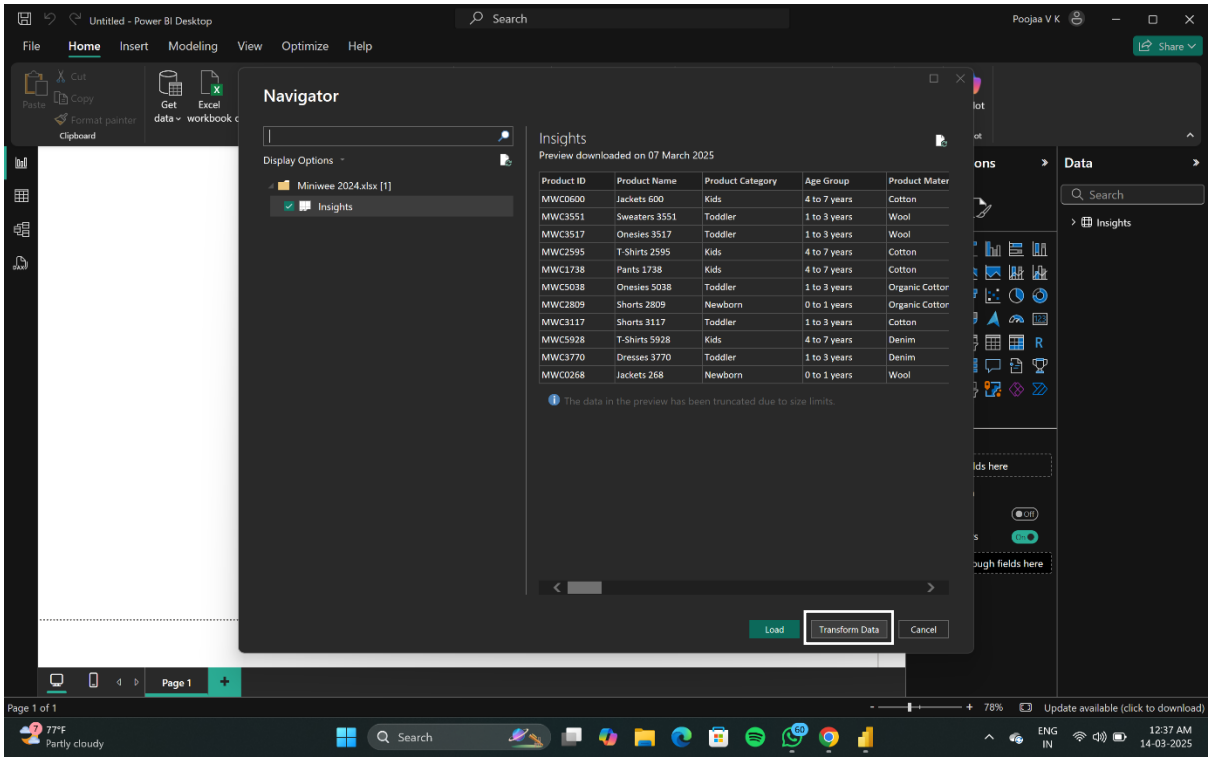

Figure 5 Previewing and Transforming Miniwee 2024 Dataset

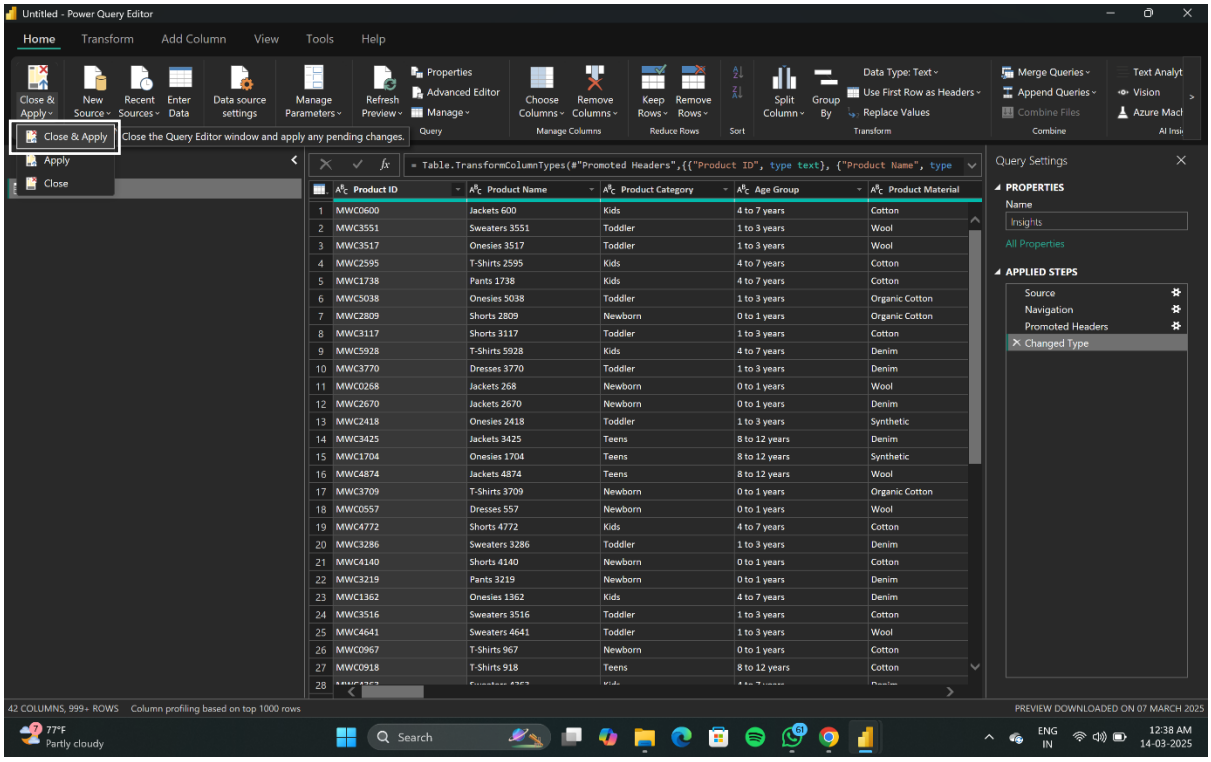

Figure 6 Refining and Cleaning Miniwee 2024 Dataset in Power BI Query Editor

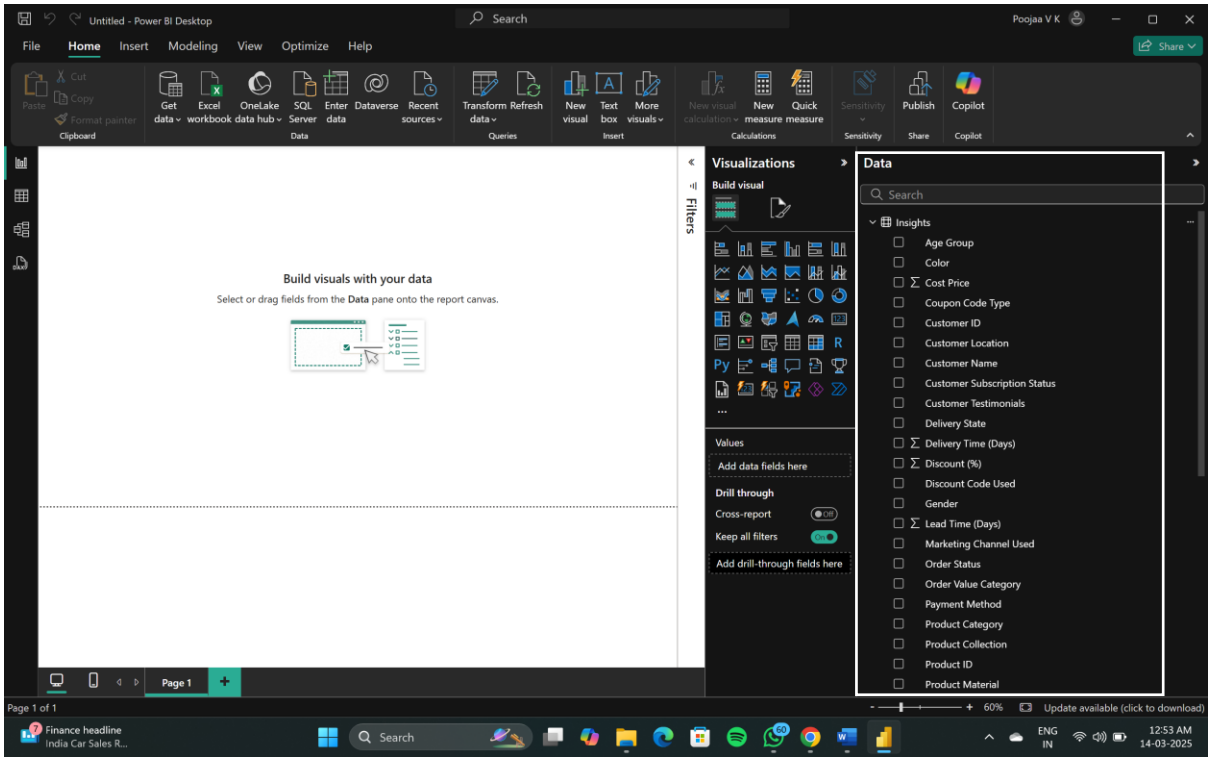

Figure 7 Building Interactive Visuals in Power BI with Miniwee 2024 Dataset

3.7.2 Output Screens

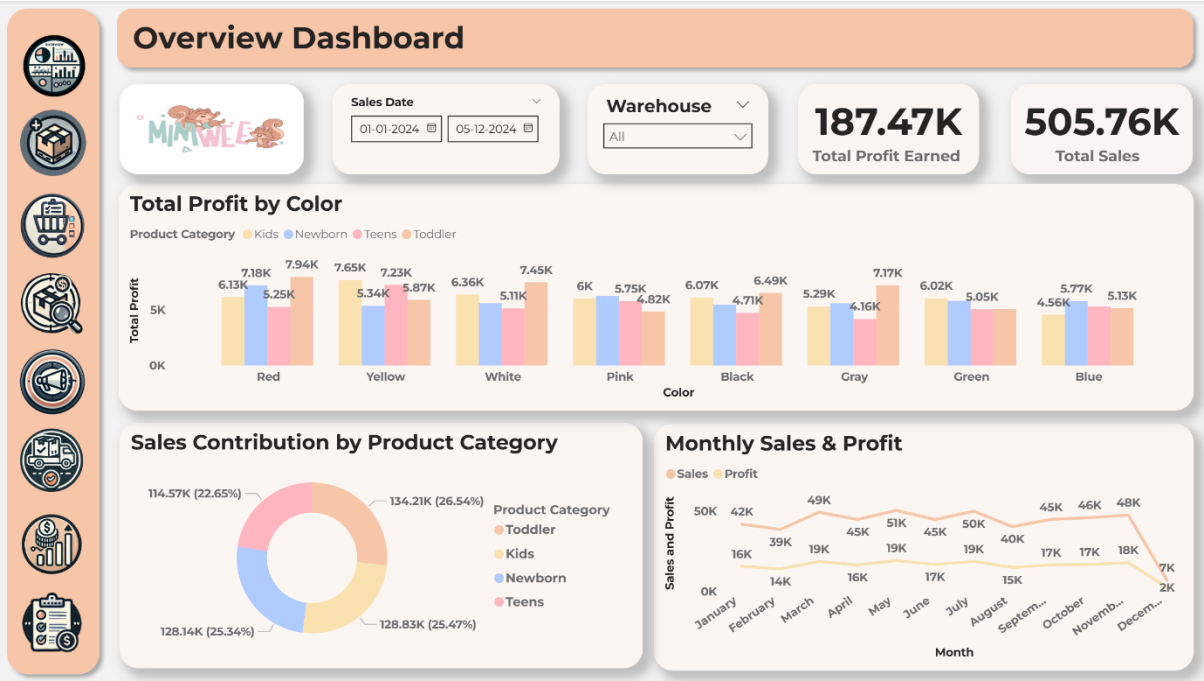

Figure 8 Overview Dashboard

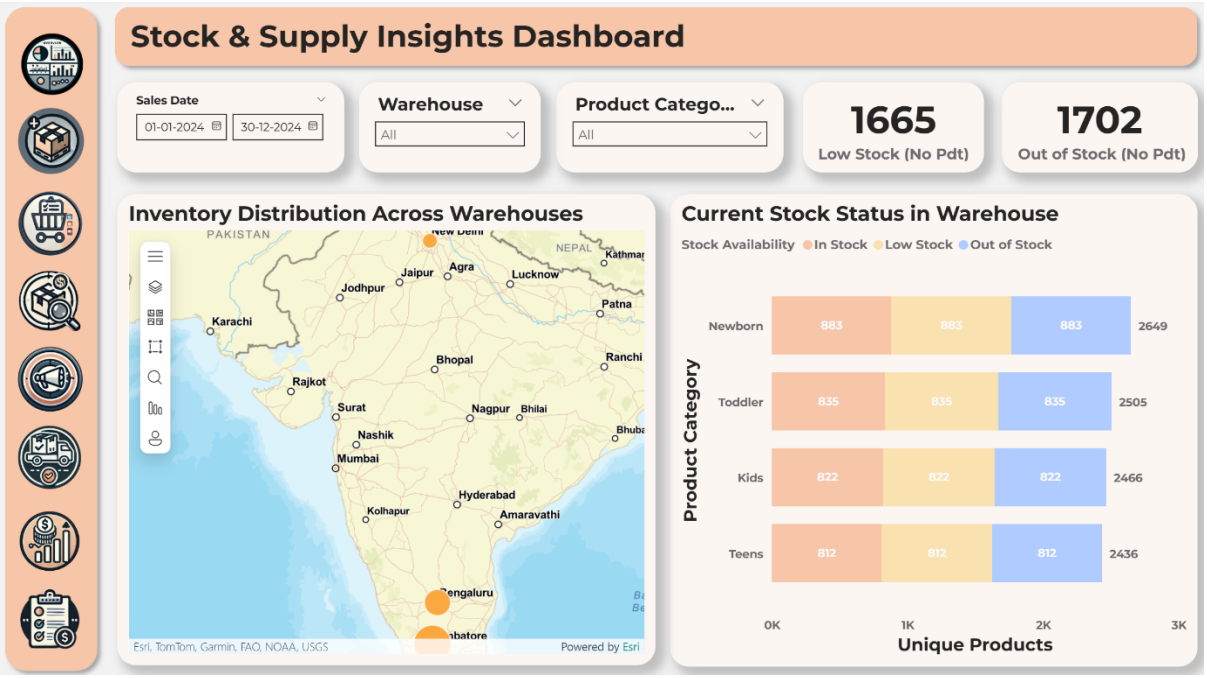

Figure 9 Stock & Supply Insights Dashboard

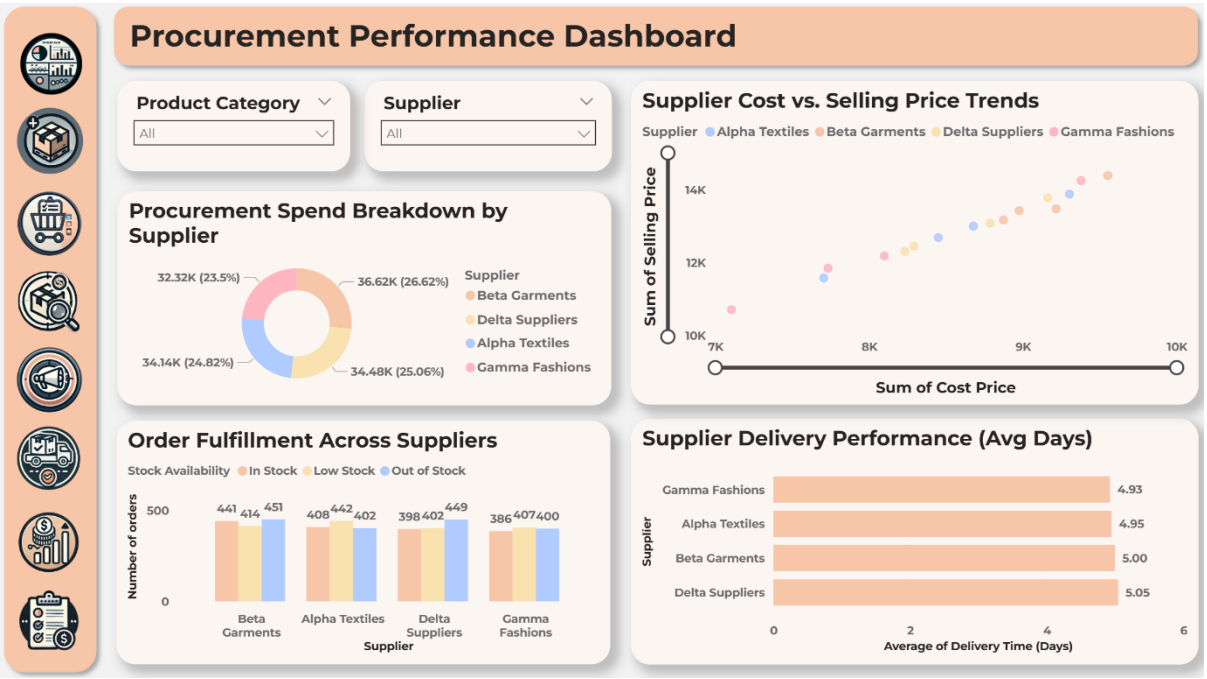

Figure 10 Procurement Performance Dashboard

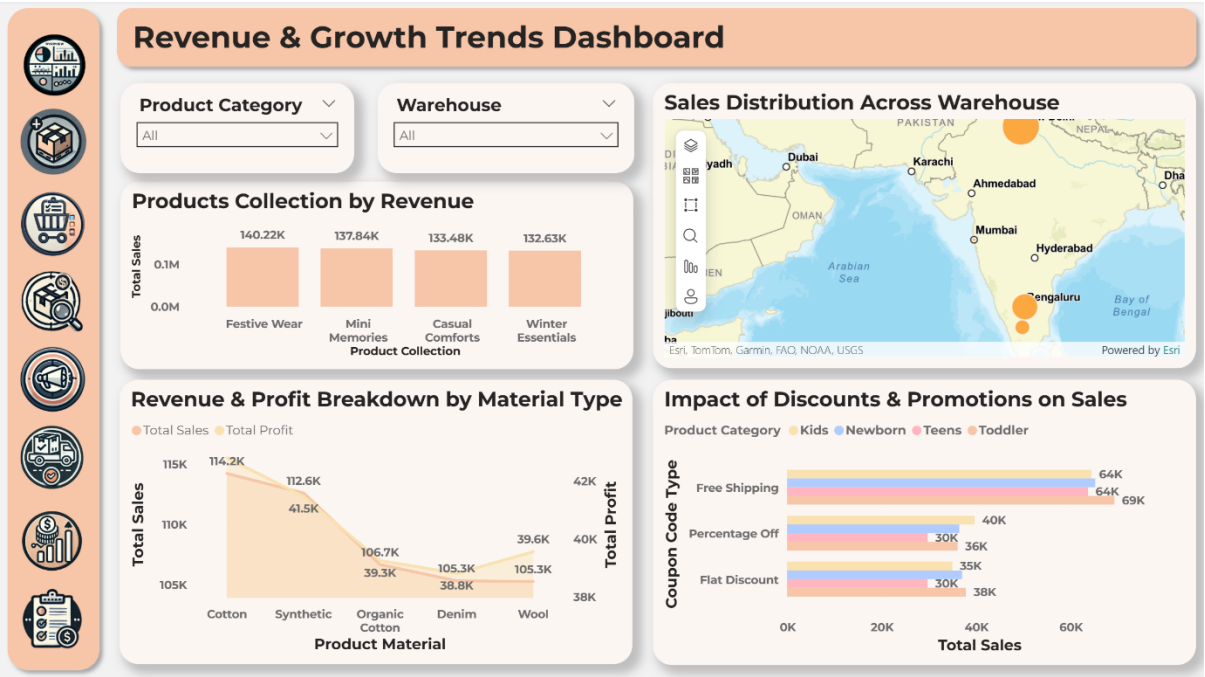

Figure 11 Revenue & Growth Trends Dashboard

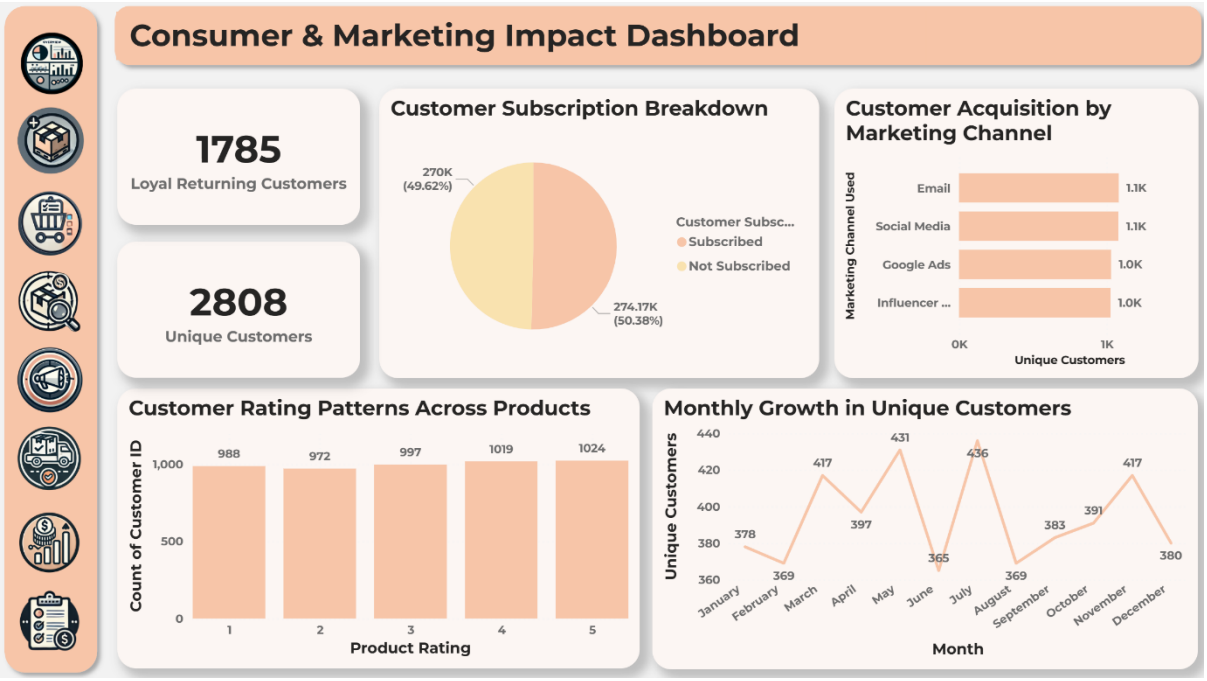

Figure 12 Consumer & Marketing Impact Dashboard

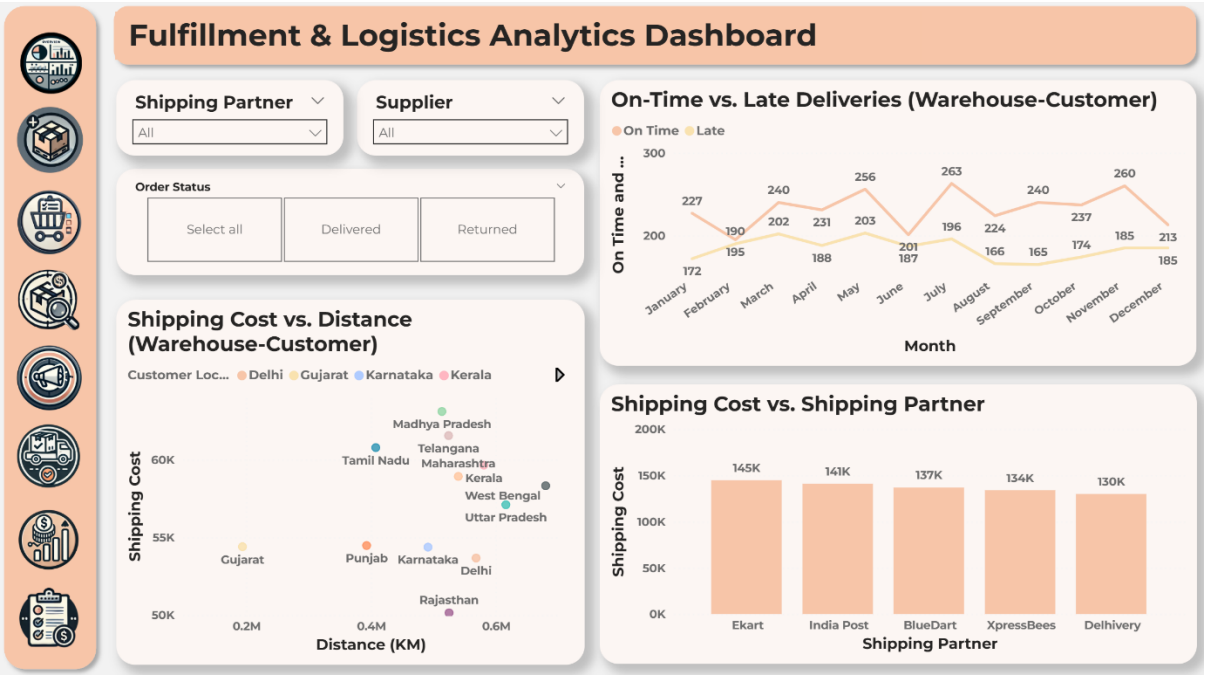

Figure 13 Fulfilment & Logistics Analytics Dashboard

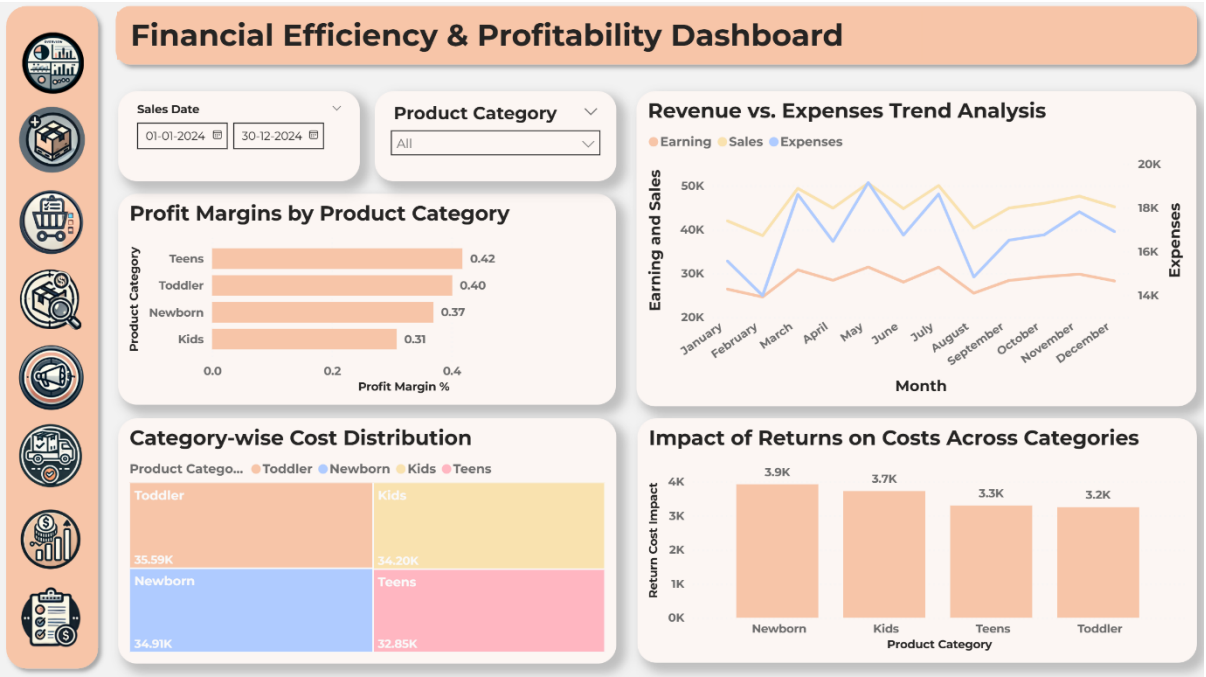

Figure 14 Financial Efficiency & Profitability Dashboard

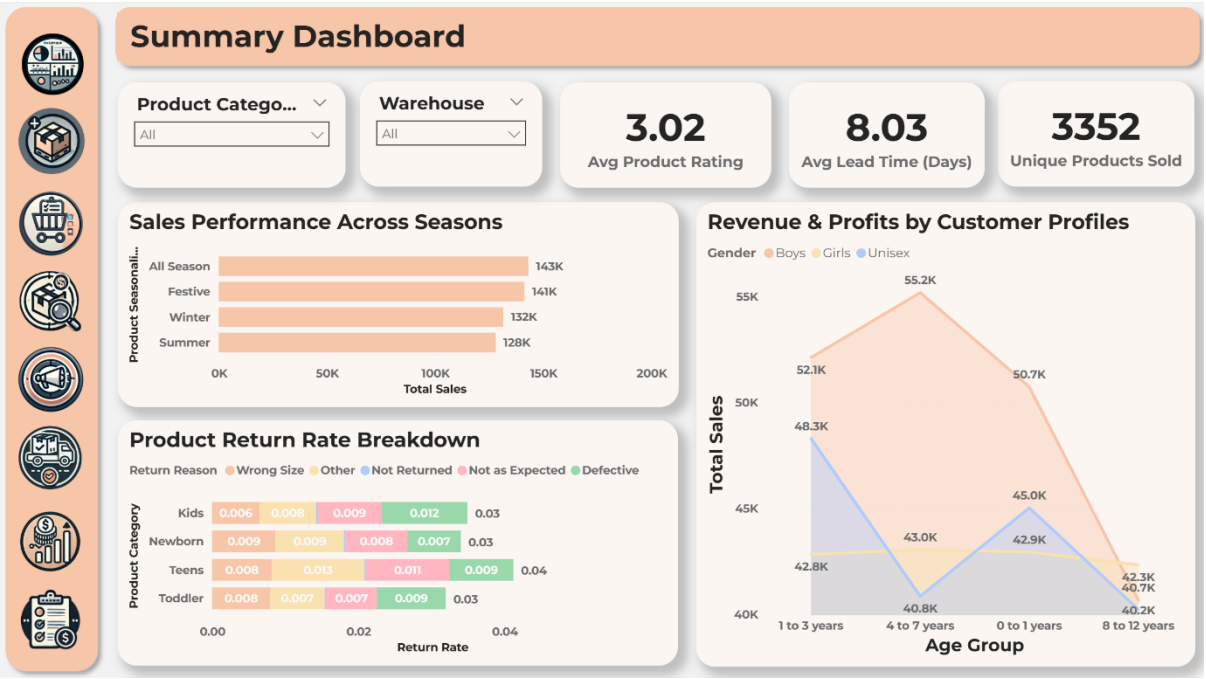

Figure 15 Summary Dashboard

IMPLEMENTATION AND RESULT

4. IMPLEMENTATION AND RESULT

The implementation of the Production and Inventory Monitoring Dashboard was carried out using Microsoft Power BI, integrating data from the Miniwee 2024 dataset. The process began with data extraction, cleaning and modelling to ensure accuracy and consistency. Key data points such as sales figures, inventory levels, delivery times, customer behaviours and supplier performance were transformed into a structured model using Power Query. Various DAX (Data Analysis Expressions) measures were created to calculate critical business metrics like actual earnings, stock turnover ratio and return rate.

The dashboard features a user-friendly interface with interactive visuals including bar charts, line charts, pie charts, matrix tables and KPI cards, allowing stakeholders to monitor performance dynamically. For instance, the stock turnover ratio visual helps assess how quickly inventory is sold, while late delivery counts highlight areas where supply chain efficiency needs improvement. The on-time delivery rate and return cost impact offer actionable insights to optimize logistics and reduce operational losses. Interactive slicers and filters enable users to drill down by product category, supplier and time period, facilitating in-depth analysis.

The results from the dashboard implementation revealed valuable business insights. It highlighted top-performing products, identified low-stock and out-of-stock items and provided a clear view of supplier performance across regions. The dashboard also improved decision-making by providing monitoring of critical KPIs. For example, return rates and customer repeat purchase patterns enabled better forecasting and customer retention strategies. Overall, the implementation of this Power BI dashboard successfully enhanced operational visibility, improved inventory management and supported data-driven decision-making, allowing for more accurate tracking and optimization of business processes.

4.1 Power BI Dashboard Development Process

The development of the Production and Inventory Monitoring Dashboard involved a structured and iterative approach to ensure accuracy, efficiency and user-friendliness. The process included several key stages: data collection, data preparation, modelling, visualization design and dashboard deployment. Each step was carefully executed to deliver insights into business performance.

4.1.1 Data Collection and Integration

The first step involved gathering data from the Miniwee 2024 dataset, which included information on sales, inventory levels, customer behaviours, delivery performance and supplier records. This dataset was imported into Power BI using Power Query, allowing for seamless integration and automated data refreshes. Multiple tables were connected using primary and foreign keys, ensuring accurate relationships between entities such as products, orders and customers.

4.1.2 Data Cleaning and Transformation

To ensure data quality, the raw dataset underwent cleaning and transformation in Power Query Editor. This included handling missing values, duplicate entries and data type conversions. Key calculations such as profit margins, return rates and delivery performance were standardized. New calculated columns and custom fields were created where required, such as stock availability categories (e.g., "In Stock," "Low Stock," "Out of Stock").

4.1.3 Data Modelling

A star schema approach was used to optimize the data model, with fact tables containing transactional data (e.g., sales and returns) and dimension tables providing descriptive context (e.g., products, customers and suppliers). Relationships between tables were defined using one-to-many cardinality. This structured model enhanced query performance and simplified the development of DAX (Data Analysis Expressions) measures.

4.1.4 Measure Development with DAX

Advanced business metrics were created using DAX functions to provide meaningful insights. Measures such as Total Sales, Actual Earnings, Stock Turnover. These measures allowed for dynamic calculations across different dimensions and time periods, enabling performance monitoring.

4.1.5 Dashboard Design and Visualization

The dashboard was designed to present information clearly and intuitively using a variety of visualization techniques. Key visuals included bar charts for category comparisons, line charts for sales trends, KPI cards for high-level metrics and matrix tables for detailed data

breakdowns. Slicers were implemented to allow users to filter by date range, product category, supplier and region, offering a customized and interactive user experience.

4.1.6 Interactivity and User Experience

Interactive features such as drill-through and tooltips were incorporated to allow users to explore detailed insights from summary views. Bookmarks were used to create navigational aids, improving ease of access to different report sections. This dynamic design enabled stakeholders to gain deeper insights and make informed decisions.

4.1.7 Testing and Deployment

Extensive testing was conducted to validate the accuracy of calculations and the functionality of interactions. Once verified, the dashboard was deployed in Power BI Service, allowing secure sharing and data updates. This implementation facilitated continuous monitoring of business operations, supporting data-driven decisions and operational improvements.

4.2 Report Publishing and Sharing

The publishing and sharing of the Production and Inventory Monitoring Dashboard in Power BI is a crucial step in making business insights accessible to stakeholders. This process ensures that the dashboard is available across different devices, allowing decision-makers to monitor key performance indicators (KPIs) and respond promptly to operational changes. The publishing workflow involves preparing the report for deployment, uploading it to the Power BI Service, configuring access permissions and enabling interactive features.

4.2.1 Preparing the Dashboard for Publishing

Before publishing, the dashboard underwent a comprehensive review to ensure data accuracy and performance optimization. Key steps included validating DAX measures, verifying data relationships and testing visual interactions like slicers, drill-throughs and tooltips. Additionally, the data model was optimized by removing unnecessary columns and ensuring efficient relationships between fact and dimension tables. The final version was saved in the .pbix format, ready for upload to the Power BI cloud service.

4.2.2 Publishing to Power BI Service

The dashboard was published directly from Power BI Desktop to the Power BI Service through a seamless integration feature. This allowed the report to be hosted on a secure, cloud-based platform, ensuring that it is accessible to authorized users. Upon publishing, the dataset and all associated measures, calculations and visuals were uploaded, enabling data interaction and scheduled data refreshes.

4.2.3 Configuring Data Refresh and Security

To ensure up-to-date insights, a scheduled data refresh was configured within the Power BI Service. This automated process updates the dashboard at regular intervals, pulling the latest data from the Miniwee 2024 dataset. Security settings were also implemented using Row-Level Security (RLS), which restricts data visibility based on user roles. For example, warehouse managers can view inventory data, while financial analysts can access earnings and cost metrics.

4.2.4 Sharing and Access Control

The dashboard was shared with stakeholders through Power BI workspaces. Role-specific access was granted, ensuring that each user group could only view the data relevant to their responsibilities. Direct sharing links and email subscriptions were also enabled, allowing stakeholders to receive automated reports and track critical metrics without needing to log into the dashboard. This method of controlled sharing helps maintain data confidentiality while promoting collaborative decision-making.

4.2.5 Mobile Accessibility and Collaboration

To enhance accessibility, the dashboard was optimized for Power BI Mobile, allowing users to access it on smartphones and tablets. This mobile compatibility ensures monitoring even outside traditional office environments. Collaboration features such as commenting and exporting to PowerPoint or PDF were also enabled, facilitating discussions and presentations.

4.3 Performance Evaluation

The Production and Inventory Monitoring Dashboard comprises multiple dashboards that analyse key business areas, including sales performance, inventory management, supplier efficiency, customer behaviour, logistics operations and profitability. Each dashboard delivers actionable insights that enable stakeholders to make informed decisions and optimize operations. Here is a detailed analysis and key insights from each dashboard:

4.3.1 Sales Performance Dashboard

The Sales Performance Dashboard tracks monthly sales and profit, providing a clear view of revenue trends. The dashboard shows that May, July and October are the most profitable months, indicating seasonal demand spikes.

Insights

- Total Sales reached 544.18K INR, with 201.92K INR in total profit
- Newborn (26.08%) and Toddler (25.56%) categories contribute the most to sales, while Teens (23.77%) lag behind
- Dresses and Jackets are the highest-profit products, while Onesies and Sweaters are least profitable, suggesting a need to reevaluate pricing or discontinue underperforming items
- Percentage-Off Promotions generate the highest sales, making them the most effective discount strategy

4.3.2 Inventory Management Dashboard

This dashboard monitors stock availability, low-stock alerts and out-of-stock items across different product categories.

Insights

- 1,702 products are currently out of stock and 1,665 products are low stock, with Kids and Toddler categories being the most affected

- Stock turnover analysis shows that slower-moving categories may require price reductions or marketing incentives to increase sales
- Regular monitoring of low-stock products can prevent supply chain disruptions and lost sales opportunities

4.3.3 Supplier Performance Dashboard

The Supplier Performance Dashboard evaluates procurement spending, order fulfilment and delivery efficiency for key suppliers.

Insights

- Beta Garments is the largest supplier, contributing 26.62% of procurement spending, followed closely by Delta Suppliers (25.06%)
- Delta Suppliers has the longest average delivery time (5.05 days), which may impact inventory availability and customer satisfaction
- Alpha Textiles consistently meets delivery targets, making them the most reliable supplier
- Monitoring supplier performance allows for better negotiations and improved supply chain efficiency

4.3.4 Customer Behaviour Dashboard

This dashboard analyses customer acquisition, subscription status and customer loyalty patterns.

Insights

- 50.38% of customers are subscribed, providing a steady revenue stream
- Email marketing is the most effective customer acquisition channel, contributing 1,082 unique customers, followed closely by social media and influencer marketing
- The dashboard shows 1,785 loyal returning customers, highlighting the importance of customer retention efforts

4.3.5 Logistics and Delivery Dashboard

The Logistics Dashboard assesses shipping costs, on-time delivery performance and partner efficiency.

Insights

- Maharashtra and Tamil Nadu incur the highest shipping costs, reflecting increased transportation expenses
- Ekart (145K INR) and India Post (141K INR) are the most expensive shipping partners
- Blue Dart and Xpress Bees offer more cost-effective delivery options while maintaining reliable delivery times
- Delivery delays are most frequent in July and October, suggesting seasonal logistics challenges

4.3.6 Profitability and Cost Analysis Dashboard

This dashboard evaluates revenue, expenses and profit margins across product categories and regions.

Insights

- Toddler (222%) and Teens (189%) categories show the highest profit margins, while Newborn (68%) has the lowest
- The Return Cost Impact reveals that Newborn (3.8K INR) incurs the highest return costs, followed by Kids (3.6K INR) and Toddler (3.6K INR)
- Managing returns and monitoring category-wise profitability helps maintain financial stability and improve cost control

CONCLUSION

5. CONCLUSION

The Production and Inventory Monitoring Dashboard developed using Power BI successfully delivers a comprehensive and interactive platform for tracking and analysing key business operations. This project addresses critical areas such as sales performance, inventory management, supplier efficiency, customer behaviour and logistics operations, providing stakeholders with accurate, insights to support data-driven decision-making. Through the integration of the Miniwee 2024 dataset, advanced DAX calculations and dynamic visualizations, the dashboard enables users to monitor essential metrics such as Total Sales, Profit Margins, Stock Availability and Supplier Performance. It offers an in-depth view of monthly and seasonal trends, allowing stakeholders to identify high-performing products and optimize inventory to prevent stockouts. The implementation of time intelligence functions enhances the dashboard's ability to track historical patterns and forecast future outcomes, facilitating better resource allocation and planning. The project overcame several technical challenges, including data integration issues, performance optimization, complex DAX calculations and ensuring data security. By optimizing the data model through a star schema and using efficient DAX patterns, the dashboard achieved faster query execution and seamless user interaction.

A key feature of the dashboard is its interactivity, allowing users to filter data by product category, warehouse location, supplier and date range. Drill-through capabilities and What-If Analysis provide additional insights, enabling scenario modelling to evaluate the impact of operational changes. This empowers decision-makers to test strategies and predict future outcomes effectively. The insights derived from the dashboard reveal crucial business patterns. For instance, the Newborn and Toddler categories consistently drive the highest revenue, while Delta Suppliers exhibit the longest delivery times, highlighting areas for improvement. By continuously monitoring customer behaviour, the business can optimize marketing strategies, improve customer retention and manage logistics costs effectively. The Production and Inventory Monitoring Dashboard enhances operational transparency, improves decision-making accuracy and supports strategic planning. It provides a scalable framework for analysing and optimizing business processes, ensuring sustainable growth and improved performance in a competitive market environment.

FUTURE ENHANCEMENT

FUTURE ENHANCEMENT

The Production and Inventory Monitoring Dashboard offers detailed insights into sales performance, inventory management, supplier efficiency, customer behaviour and logistics operations. While the current system meets the business's analytical needs, there are several areas for enhancement to further improve data accuracy, user engagement, forecasting capabilities and operational efficiency. These enhancements will make the dashboard more adaptable to evolving business requirements and emerging technologies.

AI-Powered Predictive Analytics

Incorporating AI-driven predictive analytics will enable the dashboard to provide deeper insights and more accurate forecasts. Machine learning models can be used to predict future sales trends, customer purchasing patterns and supplier lead times. For example, implementing Power BI's integration with Azure Machine Learning will allow stakeholders to forecast seasonal demand, identify risk-prone suppliers and plan inventory restocking more effectively.

Expansion of What-If Analysis

The What-If Analysis feature can be expanded to model more complex scenarios. Future enhancements can include simulations for:

- Supplier Cost Fluctuations - Evaluating the impact of increasing or decreasing procurement costs on profitability
- Market Demand Changes - Analysing how seasonal shifts or unexpected spikes in demand affect inventory levels and logistical costs
- Operational Risks - Simulating the effect of supplier delays and increasing shipping costs to assess business resilience. These advanced simulations will allow executives to make better strategic decisions and prepare for multiple business scenarios

Automated Alerts and Notifications

Implementing automated alerts for critical KPIs ensures stakeholders receive immediate notifications when performance thresholds are met. For example:

- Low-Stock Alerts - Notify inventory managers when stock falls below a predefined level
- Late Delivery Alerts - Send alerts when a supplier exceeds the expected delivery time
- Profit Margin Drops - Trigger notifications if profit margins decline beyond acceptable limits

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